HARDWARE × MODELS × REAL-WORLD SPEED

Intelligence. On your desk.

A field guide to capable local coding and tool-using agents.
14 hardware configurations. Seven local models + two cloud ceilings.

Start with the model.
Then choose memory, speed and budget.
Practical local-agent setups, from a laptop to a deskside supercomputer.

THE PORTABLE PICK
M5 Max · 128 GB
Flash-Next + MTPLX: **64 tok/s reported** at 65K context. Its n-gram table streams from SSD.

THE CAPABLE DESK-SIDE PICK
M5 Ultra · 256 GB
GLM-5.3-Flash for capability. Flash-Next for speed: 108 tok/s on short prose in oMLX.

THE COMPACT CUDA PICK
DGX Spark · 128 GB
Flash-Next at **37–46 tok/s reported**. Two Sparks have capacity for higher-scoring GLM-5.3-Flash Q4.

01 / The hardware-to-model map

M reported measurement · E estimate · — unverified / unpublished

35 tok/s gate: choose the highest-scoring fitting model whose decode range reaches ≥35 tok/s, then sort by score and speed ↑. One active request; a qualifying upper bound is not a guaranteed minimum.

DEVICE / CONFIGURATION	APPROX. USD	MEMORY INSTALLED	BANDWIDTH GB/S	AI COMPUTE TFLOPS / TOPS	BEST-FIT SHORTLIST MODEL / PRECISION	DECODE TOKENS / SEC
MacBook Pro · M5 [H1] 10-core GPU	**$2.5k** System	**32 GB** Unified	153	**~40 TOPS** E Neural Engine · est.	Qwen3.6-35B-A3B · Q4 Short context · roofline estimate	**30–50** E Short context · roofline estimate
MacBook Pro · M5 Pro [H1] 20-core GPU	**$3.7k** System	**64 GB** Unified	307	**~40 TOPS** E Neural Engine · est.	Qwen3.8-27B · dynamic Q4 MTPLX · native MTP drafting	**21–44** E Upper end: repetitive editing
Ryzen AI Max+ 395 / Halo [H2] Radeon 8060S	**$2–4k** System est.	**128 GB** GPU allocation varies	256 LPDDR5X-8000	**50 TOPS** NPU · INT8	Qwen3.8-27B · ROCmFP4_FAST Near-greedy MTP · short prompt	**30–36** M 512-token peak · B2
Radeon AI PRO R9700 [H4]	**$1.7k** Card	**32 GB**	640	**766 TOPS** INT4 · 383 TFLOPS FP8	Qwen3.8-27B · Q4 + MTP Compatible AMD drafting backend required	**30–50** E Conditional drafting · low confidence
GeForce RTX 4090 [H3]	**$2–2.5k** Card est.	**24 GB**	1,008	**661 TFLOPS** FP8	Qwen3.8-27B · Q4 Short context	**38–50** E No speculation
GeForce RTX 5090 [H5]	**$6.4–7.4k** Card est.	**32 GB**	1,792	**1,676 TFLOPS** FP4	Qwen3.8-27B · Q4_K_M LM Studio / CUDA	**59** M 6.1K · no MTP · B3
1 × DGX Spark [H7] Single NVIDIA GB10 system	**$4.7k** System	**128 GB**	273	**500 TFLOPS** FP4	Qwen3.8-Flash-Next · NVFP4 vLLM + MTP3 · table on SSD	**37–46** M Prose → code · B5
MacBook Pro · M5 Max [H1] 40-core GPU	**$6k** System est.	**128 GB** Unified	614	**~40 TOPS** E Neural Engine · est.	Qwen3.8-Flash-Next · dynamic Q4 MTPLX 2.12 + MTP · n-gram table on SSD	**63.5** M 65.5K prompt · B1
RTX PRO 6000 Blackwell [H6] Workstation Edition	**$16–20k** Card	**96 GB**	1,792	**2,015 TFLOPS** FP4	Qwen3.8-Flash-Next · NVFP4 SSD lookup table · fit projected	**65–125** E Best ~95 · roofline estimate
2 × DGX Spark [H7]	**$9.4k** Systems	**256 GB** 128 GB per node	2 × 273 Distributed memory	**1,000 TFLOPS** FP4 · summed	GLM-5.3-Flash · IQ4_XS TP2 · ~157 GB weights · fit projected	**20–40** E Drafting scenario · low confidence
Mac Studio · M5 Ultra [H8] 80-core GPU	**$12k** System est.	**256 GB**	1,200	**~80 TOPS** E Neural Engine · est.	GLM-5.3-Flash · mixed 4/8-bit oMLX	**28–41** M 61K → short · B3
M5 Ultra · 512 GB [H8] Unavailable · expected late October	**$15–18k** System est.	**512 GB**	1,200	**~80 TOPS** E Neural Engine · est.	GLM-5.3-Flash · mixed 4/8-bit Same-chip 256 GB result extrapolated	**28–41** E Same bandwidth · more headroom
2 × RTX PRO 6000 Blackwell [H6]	**$32–40k** Cards est.	**192 GB** 96 GB per GPU	2 × 1,792 Not one shared bus	**4,030 TFLOPS** FP4 · summed	GLM-5.3-Flash · IQ4_XS TP2 · ~157 GB weights · fit projected	**60–110** E Best ~80 · roofline estimate
DGX Station · GB300 [H9] OEM order · deskside	**$99–124k** System	**748 GB** 252 HBM + 496 CPU	7,100 / 396 HBM / LPDDR5X	**15,000 TFLOPS** FP4 vendor	GLM-5.3 · IQ4_XS, split HBM / CPU Conditional drafting-enabled setup	**22–45** E Drafting assumed · low confidence

Prices: approximate USD before tax; cards exclude the host PC; "est." means a budget estimate. **Speeds:** externally reported (M) or estimated (E); model, context and runtime differ. **Compute:** GPU dense peaks, or NPU/Neural Engine throughput where labeled; Apple E values are rough scale estimates, not rated peaks or total-chip compute.

02 / Local models & frontier ceilings

AA Intelligence Index v4.3.2 · higher is better

10 independent evaluations: agents 30% · coding 20% · scientific reasoning 20% · general 30%. A shared capability composite with substantial agentic coverage.

MODEL / TESTED SETTING	AA SCORE	PARAMETERS / TYPE	MEMORY ESTIMATE · 128K CONTEXT · 1 ACTIVE REQUEST
Claude Opus 5.5 Max · default fallback · cloud	**58**	**Undisclosed** Architecture undisclosed	**Cloud only · weights and serving memory undisclosed**
GPT-6 Astra Max · cloud ceiling	**53**	**Undisclosed** Architecture undisclosed	**Cloud only · weights and serving memory undisclosed**
GLM-5.3 Max	**45**	**~753B · MoE** ~40B active	**365 GB weights + ~13 GB cache ≈ 378 GB resident** IQ4_XS · compressed MLA, 16-bit cache · SSD tables: none · runtime: +8–16 GB
GLM-5.3-Flash Reasoning	**42**	**320B · MoE** 18B active	**157–200 GB weights + ~2 GB cache ≈ 159–202 GB resident** Q4-class · compressed MLA, 16-bit cache · SSD tables: none · runtime: +8–16 GB Expanded-KV backends need ~95–105 GB cache instead.
Qwen3.8-Flash-Next Reasoning	**40**	**~180B · MoE + lookup** 125B backbone · 6B active	**~83 GB weights + ~4 GB cache ≈ 87 GB resident** MTPLX dynamic Q4 · 16-bit cache · +32 GB lookup tables on SSD · runtime: +4–8 GB
DeepSeek V4.1 Flash Max	**39**	**~763B · MoE + Engram** 552B backbone · 16B active decode	**~316 GB weights + ~0.2–0.6 GB cache ≈ 317 GB resident** DwarfStar Q4 · packed FP4–BF16 cache scenarios · +203 GB Engram on SSD · runtime: +8–16 GB
Qwen3.8-27B Xhigh	**34**	**27B · Dense** 27B active	**20.4 GB weights + ~9.3 GB cache ≈ 30 GB resident** MTPLX dynamic Q4 · 16-bit cache · SSD tables: none · runtime: +4–8 GB
DeepSeek V4 Flash / 0731 Max · text model	**34**	**284B · MoE** 13B active	**~155 GB weights + ~1 GB cache ≈ 156 GB resident** Q4_K_XL · compressed attention, BF16 cache · SSD tables: none · runtime: +8–16 GB
Qwen3.6-35B-A3B Reasoning · fallback	**18**	**35B · MoE** ~3B active	**~22 GB weights + ~3 GB cache ≈ 25 GB resident** Q4 · 16-bit cache · runtime: +4–8 GB